

Supplementary Materials for

The HIV Post-Exposure Prophylaxis Window: A Multiscale Framework Linking Within-Host Integration Kinetics to Population-Scale Structural Access

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This PDF file includes:

- Sections S1–S6: Mathematical development with complete proofs
- Section S7: Multiscale within-host model implementation
- Section S8: NHP concordance — per-timepoint detail
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S1 Probability Space, State Definitions, and Assumptions

Consider a single HIV exposure event e occurring at time $t = 0$ on probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $\{\mathcal{F}_t\}_{t \geq 0}$.

S1.1 Within-host states

Definition S1.1 (Within-host infection states). $Z(t) \in \{0, 1, 2\}$:

- $Z(t) = 0$ (Susceptible): no productive infection; $I(t) < I^*$.
- $Z(t) = 1$ (Seeded): replicative focus established ($I(t) \geq I^*$) but reservoir below irreversibility threshold ($R(t) < R^*$).
- $Z(t) = 2$ (Integrated): $R(t) \geq R^*$. Absorbing state under Assumption A1.

Definition S1.2 (Transition stopping times).

$$T_{\text{seed}} = \inf\{t \geq 0 : Z(t) \geq 1\}, \quad T_{\text{int}} = \inf\{t \geq 0 : Z(t) = 2\}.$$

We assume $T_{\text{seed}} \leq T_{\text{int}}$ almost surely.

Definition S1.3 (Cumulative transition probabilities). $P_{\text{seed}}(t) = \mathbb{P}(T_{\text{seed}} \leq t)$, $P_{\text{int}}(t) = \mathbb{P}(T_{\text{int}} \leq t)$. Since $T_{\text{seed}} \leq T_{\text{int}}$ a.s., $P_{\text{seed}}(t) \geq P_{\text{int}}(t)$ for all $t \geq 0$.

S1.2 Assumptions

Assumption S1.1 (Irreversibility of integration). Once $Z(t) = 2$, the state is absorbing: $\mathbb{P}(Z(s) = 0 \mid Z(t) = 2) = 0$ for all $s > t$.

Biological grounding: Integrated provirus persists for the lifetime of the host cell and propagates to daughter cells via clonal expansion. ART suppresses replication ($I \rightarrow 0$) but $dR/dt = \alpha I \rightarrow 0$: the reservoir is static under ART but does not shrink. No approved therapy achieves eradication.

Assumption S1.2 (Stage-dependent drug efficacy).

$$\varepsilon(Z(t)) = \begin{cases} \varepsilon_{\max} & Z(t) = 0, \\ \varepsilon_{\text{mid}} & Z(t) = 1, \\ \varepsilon_{\min} = 0 & Z(t) = 2, \end{cases} \quad \text{with } 1 \geq \varepsilon_{\max} \geq \varepsilon_{\text{mid}} \geq 0.$$

Assumption S1.3 (Monotonicity of integration). $P_{\text{int}}(t)$ is non-decreasing, $P_{\text{int}}(0) = 0$, $\lim_{t \rightarrow \infty} P_{\text{int}}(t) = 1$.

Assumption S1.4 (Transmission-competent exposure). Source plasma viral load is at or above the $U=U$ threshold (≥ 200 copies/mL sustained). The framework is silent on exposures from virologically suppressed sources, for whom transmission risk is functionally zero (Rodger 2019, PARTNER2; Bavinton 2018, Opposites Attract).

S2 Master Equation

Proposition S2.1 (Master equation). *Under Assumption A2,*

$$E_{\text{PEP}}(t) = (1 - P_{\text{seed}}(t)) \varepsilon_{\max} + (P_{\text{seed}}(t) - P_{\text{int}}(t)) \varepsilon_{\text{mid}} + P_{\text{int}}(t) \varepsilon_{\min} \quad (\text{S1})$$

Proof. Partition the sample space at time t by $Z(t)$:

$$\mathbb{P}(Z = 0) = 1 - P_{\text{seed}}, \quad \mathbb{P}(Z = 1) = P_{\text{seed}} - P_{\text{int}}, \quad \mathbb{P}(Z = 2) = P_{\text{int}}.$$

These events are mutually exclusive and exhaustive. By the law of total expectation:

$$E_{\text{PEP}}(t) = \mathbb{E}[\varepsilon(Z(t))] = (1 - P_{\text{seed}}) \varepsilon_{\max} + (P_{\text{seed}} - P_{\text{int}}) \varepsilon_{\text{mid}} + P_{\text{int}} \varepsilon_{\min}. \quad \square$$

S3 The Finite Prevention Window Theorem

Theorem S3.1 (Finite Prevention Window). *Under Assumptions A1–A3, $E_{\text{PEP}}(t)$ satisfies:*

- (i) $E_{\text{PEP}}(t) \leq (1 - P_{\text{int}}(t)) \varepsilon_{\max}$;
- (ii) $E_{\text{PEP}}(t)$ is non-increasing in t ;

(iii) $E_{PEP}(t) \rightarrow 0$ as $t \rightarrow \infty$;

(iv) for any $\eta \in (0, 1)$, $t_{crit}(\eta) = \inf\{t : E_{PEP}(t) < \eta\}$ is finite.

Proof. Part (i) — Upper bound. Replace ϵ_{mid} with ϵ_{max} in (S1):

$$E_{PEP}(t) \leq (1 - P_{seed})\epsilon_{max} + (P_{seed} - P_{int})\epsilon_{max} + P_{int}\epsilon_{min} = (1 - P_{int})\epsilon_{max} + P_{int}\epsilon_{min}.$$

With $\epsilon_{min} = 0$: $E_{PEP}(t) \leq (1 - P_{int}(t))\epsilon_{max}$.

Part (ii) — Monotone decay. For $t_1 < t_2$, let $\Delta_S = P_{seed}(t_2) - P_{seed}(t_1) \geq 0$ and $\Delta_I = P_{int}(t_2) - P_{int}(t_1) \geq 0$.

Then

$$E_{PEP}(t_1) - E_{PEP}(t_2) = \Delta_S(\epsilon_{max} - \epsilon_{mid}) + \Delta_I(\epsilon_{mid} - \epsilon_{min}) \geq 0.$$

Part (iii) — Convergence. By A3, $P_{int}(t) \rightarrow 1$. Since $T_{seed} \leq T_{int}$ a.s., $P_{seed}(t) \rightarrow 1$. Limits give $E_{PEP}(t) \rightarrow 0 \cdot \epsilon_{max} + 0 \cdot \epsilon_{mid} + 1 \cdot \epsilon_{min} = 0$.

Part (iv) — Finite critical time. By (ii), E_{PEP} is non-increasing; by (iii), $E_{PEP} \rightarrow 0 < \eta$. Therefore $\{t : E_{PEP}(t) < \eta\}$ is non-empty, and $t_{crit}(\eta)$ is finite. $\square$

S4 Hazard-Rate Decomposition of Efficacy Loss

Definition S4.1 (Hazard rates). $\lambda_{seed}(t) = f_{seed}(t)/(1 - P_{seed}(t))$, $\lambda_{int}(t) = f_{int}(t)/(P_{seed}(t) - P_{int}(t))$.

Proposition S4.1 (Decomposition). Under Assumptions A1–A3 with absolute continuity,

$$\frac{dE_{PEP}}{dt} = -f_{seed}(t)\Delta\epsilon_1 - f_{int}(t)\Delta\epsilon_2 \leq 0 \tag{S2}$$

where $\Delta\epsilon_1 = \epsilon_{max} - \epsilon_{mid}$ and $\Delta\epsilon_2 = \epsilon_{mid} - \epsilon_{min}$.

Proof. Differentiating (S1):

$$\frac{dE_{PEP}}{dt} = -f_{seed}\epsilon_{max} + (f_{seed} - f_{int})\epsilon_{mid} + f_{int}\epsilon_{min} = -f_{seed}\Delta\epsilon_1 - f_{int}\Delta\epsilon_2 \leq 0. \quad \square$$

Remark. Equation S2 decomposes efficacy loss into a seeding channel (f_{seed} , gap $\Delta\epsilon_1$) and an integration channel (f_{int} , gap $\Delta\epsilon_2$). At early times, seeding dominates; as mass accumulates in state 1, the integration channel accelerates terminal efficacy collapse.

S5 Route Compression and Inoculum-Monotone Hitting Times

S5.1 ODE system

The standard target-cell-limited within-host model:

$$\frac{dT}{dt} = \lambda - dT - \beta TV, \quad \frac{dI}{dt} = \beta TV - \delta I, \quad \frac{dV}{dt} = pI - cV, \quad \frac{dR}{dt} = \alpha I \quad (\text{S3})$$

where T = target cells, I = infected cells, V = free virions, R = cells with stable proviral integration.

Lemma S5.1 (Absorbing state). *The set $\mathcal{A} = \{R > 0\}$ is positively invariant. If $R(t_0) > 0$ then $R(t) > 0$ for all $t > t_0$.*

Proof. $dR/dt = \alpha I \geq 0$, so R is non-decreasing. Under ART, $I \rightarrow 0$ and $dR/dt \rightarrow 0$: the reservoir is static but does not shrink. $\square$

S5.2 Route-dependent initial conditions

$V^{(m)}(0) = 1$ virion/mL (mucosal, post-epithelial attenuation) versus $V^{(p)}(0) = 10^3$ virions/mL (parenteral, direct intravascular delivery). All ODE parameters in (S3) are identical across routes.

Lemma S5.2 (Inoculum-monotone hitting times). *Solutions of (S3) with $V(0) = v_0 < \tilde{V}(0) = \tilde{v}$ satisfy $\tilde{I}(t) \geq I(t)$ and $\tilde{R}(t) \geq R(t)$ for all $t \geq 0$. Therefore $T_{\text{int}}(\tilde{v}) \leq T_{\text{int}}(v_0)$ a.s.*

Proof. The subsystem (I, V, R) is cooperative for fixed T :

$$\frac{\partial(\beta TV)}{\partial V} \geq 0, \quad \frac{\partial(pI)}{\partial I} \geq 0, \quad \frac{\partial(\alpha I)}{\partial R} = 0.$$

By comparison theorems for cooperative ODE systems[7], solutions are order-preserving in initial conditions. Since $\tilde{V}(0) > V(0)$: $\tilde{V}(t) \geq V(t)$ and $\tilde{I}(t) \geq I(t)$ for all $t \geq 0$. Then

$$\tilde{R}(t) = \alpha \int_0^t \tilde{I}(s) ds \geq \alpha \int_0^t I(s) ds = R(t),$$

so $T_{\text{int}}(\tilde{v}) \leq T_{\text{int}}(v_0)$. $\square$

S5.3 Route compression corollary

Assumption S5.1 (Stochastic dominance). $P_{\text{int}}^{(p)}(t) \geq P_{\text{int}}^{(m)}(t)$ for all $t \geq 0$. Follows from Lemma S5.2 under route-specific inocula.

Corollary S5.1 (Route compression). *Under Assumptions A1–A4:*

- (i) $E_{PEP}^{(p)}(t) \leq E_{PEP}^{(m)}(t)$;

$$(ii) \ t_{crit}^{(p)}(\eta) \leq t_{crit}^{(m)}(\eta).$$

Proof. (i) The weight on $\epsilon_{\max}$ equals $1 - P_{\text{seed}}(t)$, which is smaller for parenteral; the weight on $\epsilon_{\min} = 0$ equals $P_{\text{int}}(t)$, which is larger for parenteral. So $E_{\text{PEP}}^{(p)} \leq E_{\text{PEP}}^{(m)}$.

(ii) Both functions are non-increasing; $t_{crit}^{(p)}(\eta) = \inf\{t : E_{\text{PEP}}^{(p)} < \eta\} \leq \inf\{t : E_{\text{PEP}}^{(m)} < \eta\} = t_{crit}^{(m)}(\eta)$. $\square$

S6 Population-Level PEP Efficacy Bound

Definition S6.1. $\bar{E}_{\text{PEP}} = \mathbb{E}[E_{\text{PEP}}(T_{\text{access}})] = \int_0^\infty E_{\text{PEP}}(t) dF_{\text{access}}(t)$.

Proposition S6.1 (Population-level bound). *Under Assumptions A1–A4,*

$$\bar{E}_{\text{PEP}} \leq F_{\text{access}}(t_{crit}) \epsilon_{\max} + (1 - F_{\text{access}}(t_{crit})) \eta \quad (\text{S4})$$

Proof. Partition the integral at $t_{crit}(\eta)$:

$$\bar{E}_{\text{PEP}} = \int_0^{t_{crit}} E_{\text{PEP}}(t) dF_{\text{access}} + \int_{t_{crit}}^\infty E_{\text{PEP}}(t) dF_{\text{access}} \leq \epsilon_{\max} F_{\text{access}}(t_{crit}) + \eta (1 - F_{\text{access}}(t_{crit})),$$

using $E_{\text{PEP}}(t) \leq \epsilon_{\max}$ for $t \leq t_{crit}$ and $E_{\text{PEP}}(t) < \eta$ for $t > t_{crit}$ (Theorem S3.1(iv)). $\square$

S6.1 Worked example (verified numerical values)

Using the multiscale model output at the canonical operating point ($V_0 = 10^3$ parenteral, $\text{CV} = 0.3$ on β, c, δ, α ; commit 37e27ea), $t_{crit}^{(p)}(0.05) \approx 34.5$ h. Modeling access delay as log-normal (median 96 h, geometric SD 2.0):

$$F_{\text{access}}(34.5 \text{ h}) = \Phi\left(\frac{\ln 34.5 - \ln 96}{\ln 2}\right) = \Phi(-1.476) \approx 0.0699.$$

With $\eta = 0.05$ and $\epsilon_{\max} = 0.95$:

$$\bar{E}_{\text{PEP}} \leq 0.0699 \times 0.95 + 0.9301 \times 0.05 \approx 0.113.$$

Envelope reading: expected population PEP efficacy below approximately 11.3%.

S6.2 Sensitivity of the envelope to F_{access} parameterization

The median time-to-PEP-access for PWID is not directly anchored by primary timing data in the cited references (Taylor 2019[10] is a 3-page commentary; Baugher 2025[11] is NHBS-PWID surveillance focused on PrEP/syringe metrics, not PEP timing). The median is therefore a modeling choice informed by qualitative discussion of structural barriers in those references. Sensitivity:

Table S1: Envelope sensitivity to F_{access} median (GSD = 2.0, $t_{\text{crit}}^{(p)} = 34.5$ h, $\eta = 0.05$, $\varepsilon_{\text{max}} = 0.95$).

Median	$F_{\text{access}}(34.5 \text{ h})$	Envelope $\bar{E}_{\text{PEP}}$
72 h (JID-consistent)	0.144	$\leq 18.0\%$
96 h (canonical)	0.070	$\leq 11.3\%$
110 h	0.047	$\leq 9.2\%$
120 h	0.036	$\leq 8.2\%$

The envelope bound is the most generous (least pessimistic) reading of population-level PEP efficacy under the framework; in independent cascade-based modeling of structural access (companion analysis; see main text Population-level efficacy section), point estimates of $P(R_0 = 0)$ for PWID under current US policy are 0.003–0.7%, one to four orders of magnitude below the envelope. Both lines of evidence support the qualitative claim that PEP cannot serve as a population-level safety net for PWID under current structural conditions; the envelope provides the analytical bound, the cascade evidence provides the central estimates.

S6.3 Limitation: F_{access} denominator

F_{access} derives from population-wide PEP-access surveillance, which includes exposures from both transmission-competent and virologically suppressed sources. Under U=U (Assumption A4), exposures from suppressed sources are outside the framework’s domain. Restricting F_{access} to transmission-competent exposures only would reduce the relevant denominator and yield a tighter (smaller) F_{access} and therefore a tighter envelope. The reported envelope is thus conservative with respect to U=U scoping: the true within-domain envelope is no larger than the reported value.

S7 Multiscale Within-Host Model: Implementation

The multiscale within-host model combines stochastic founder-phase dynamics with deterministic ODE integration once productive infection is established. This section documents the implementation; all code is reproducible from SRC/multiscale_model/ at commit 37e27ea of the project repository.

S7.1 Phase 1: Stochastic founder-phase tau-leaping

At low founder population ($V_0 \leq I_{\text{handoff}} = 10$), the dynamics are sensitive to extinction events; deterministic integration of (S3) systematically underestimates the extinction probability and overestimates the rate of reservoir establishment. Phase 1 uses tau-leaping integration of the discrete-state Markov chain with reactions:

$$T \xrightarrow{\beta TV} I, \quad I \xrightarrow{\delta} \emptyset, \quad I \xrightarrow{p} I + V, \quad V \xrightarrow{c} \emptyset, \quad I \xrightarrow{\alpha} R.$$

A fixed-delay buffer of $\tau_{\text{eclipse}} = 0.9$ days (≈ 22 h, Perelson 1996 Table 2)[5] is enforced between cell infection and integration competence, ensuring that the lower bound on T_{int} matches the biological eclipse

floor regardless of population dynamics.

A realization is classified as *extinct* if both $I(t)$ and $V(t)$ reach zero before the population threshold $I_{\text{handoff}} = 10$ is exceeded. A realization is classified as *integrated* if $R(t) \geq R^* = 10$ cells/mL is reached. For mucosal exposure ($V_0 = 1$), extinction probability under $\text{CV} = 0.3$ is $\approx 68\%$; for parenteral exposure ($V_0 = 10^3$, $\text{CV} = 0.3$), extinction is essentially zero ($< 0.5\%$).

S7.2 Phase 2: Deterministic ODE handoff

Once $I(t)$ exceeds $I_{\text{handoff}} = 10$, the dynamics transition to deterministic integration of (S3) using `scipy.integrate.solve_ivp` with adaptive step size. The handoff continues until $R(t) \geq R^*$, at which point T_{int} is recorded.

S7.3 Heterogeneity

Inter-individual variability is modeled as multiplicative log-normal noise on $\{\beta, c, \delta, \alpha\}$ with coefficient of variation 0.3. The noise parameters are sampled once per realization; deterministic phase parameters use the sampled values throughout. The $\text{CV} = 0.3$ value is consistent with the standard within-host model heterogeneity assumption used in earlier published noise-anchored models.

S7.4 Sample size and seeding

We use $N = 500$ realizations per (route, V_0 , CV) cell. Random seeds are deterministic: `rng = np.random.default_rng(base_seed*1,000,000+k)` where k is the realization index. This guarantees byte-identical reproducibility from a cloned repository at the cited SHA. Sensitivity to N at the canonical operating point is reported in main text Methods; t_{crit} estimates are stable to ± 0.5 h across $N \in \{200, 500, 1000, 5000\}$.

S7.5 Output: t_{crit} and $P_{\text{int}}(t)$

For each cell, the empirical CDF $P_{\text{int}}(t)$ is constructed from the N -realization integration-time distribution. The route-specific efficacy curve $E_{\text{PEP}}(t)$ is computed from $P_{\text{int}}(t)$ via the master equation (S1), and $t_{\text{crit}}(\eta)$ is the smallest t at which $E_{\text{PEP}}(t) < \eta$. Reported t_{crit} values:

Table S2: Verified t_{crit} at $\eta = 0.05$ from SRC/multiscale_model/results_v3/heterogeneity_summary.csv.

Route	V_0	CV	$t_{\text{crit}}(0.05)$ [h]
Parenteral	10^3	0.0	32.5
Parenteral	10^3	0.3	34.5
Parenteral	10^3	0.5	36.0
Mucosal	1	0.0	57.5
Mucosal	1	0.3	60.5
Mucosal	1	0.5	60.0

The compression ratio at the canonical operating point (CV = 0.3) is $60.5/34.5 \approx 1.75\times$.

S8 NHP Concordance — Per-Timepoint Detail

The multiscale model output was compared to NHP empirical protection rates at the timepoints reported in Tsai 1998[8] and Otten 2000[9]. Concordance is defined as $|E_{\text{PEP}}(\text{delay}) - \text{NHP protection}| < 0.20$ (within 20 percentage points).

Table S3: Per-timepoint NHP concordance at the canonical operating point (CV = 0.3). Reproducible from SRC/multiscale_model/results_v3/nhp_concordance.csv.

Study	Route	Delay [h]	NHP protection	Model E_{PEP}	diff [pp]	Concordant
Otten 2000	mucosal	12	1.00	0.95	5.0	yes
Otten 2000	mucosal	36	1.00	0.95	5.0	yes
Otten 2000	mucosal	72	0.50	0.016	48.4	no
Tsai 1998 IV	parenteral	24	1.00	0.50	50.0	no
Tsai 1998 IV	parenteral	48	0.50	0.00	50.0	no

Compression-ratio concordance. The model’s mucosal/parenteral t_{crit} ratio at $\eta = 0.05$ is $60.5/34.5 \approx 1.75\times$. The empirical NHP compression ratio between Tsai 1998 IV (parenteral half-protection at 48 h) and Otten 2000 (mucosal half-protection at 72 h) is $72/48 = 1.50\times$. The model’s compression value is concordant with the empirical NHP range of $1.5\text{--}2\times$.

Per-timepoint absolute discordance. The within-host kinetic model under-predicts protection at:

- Tsai 1998 IV at 24 h (model 0.50 vs. NHP 1.00),
- Tsai 1998 IV at 48 h (model 0.00 vs. NHP 0.50),
- Otten 2000 at 72 h (model 0.016 vs. NHP 0.50).

We interpret this gap as evidence of NHP-specific protective contributions not captured by the within-host kinetic framework alone. Possible mechanisms include innate immunity contributions during the eclipse phase, mucosal containment by epithelial defense, and the small NHP cohort size ($n = 2\text{--}4$ per group) producing wider empirical confidence intervals on observed protection rates than the model’s first-passage

formulation accommodates. The compression-ratio result and the absolute-timepoint result are independent: the compression result is robust to the model’s domain limitations; absolute-timepoint discordance constrains the interpretation of the model as a kinetic-only ceiling rather than a full mechanistic prediction of NHP protection.

S9 Parameter Tables

S9.1 ODE system parameters (Perelson 1996 anchored)

Table S4: ODE system parameters anchored on Perelson et al. 1996 (PMID 8599114).

Symbol	Description	Value	Units	Source
δ	Infected cell death rate	0.7	day^{-1} ($t_{1/2} \approx 1.6\text{d}$)	Perelson 1996, Table 1
c	Viral clearance rate	23	day^{-1} ($t_{1/2} = 0.24\text{d} \approx 6\text{h}$)	Perelson 1996, Table 1
τ	Viral generation time	2.6	days ($\approx 62\text{h}$)	Perelson 1996, Table 2
Eclipse phase	Min time to integration competence	0.9	days ($\approx 22\text{h}$)	Perelson 1996, Table 2
β	Infection rate constant	2.4×10^{-5}	$\text{mL} \cdot \text{virion}^{-1} \cdot \text{day}^{-1}$	Standard model
p	Virion production per infected cell	10^3	virions/cell/day	Standard model
α	Stable integration rate	10^{-3}	day^{-1}	Calibrated

Parameters c and δ directly from Table 1; eclipse phase and generation time from Table 2. *Note:* the reference $\log_{10} \text{VL} = 4.5$ for PWID is from NHBS/NHAS surveillance, not from Perelson 1996 (whose patients had mean $\log_{10} \text{VL} \approx 5.3$).

S9.2 Drug efficacy parameters

Table S5: Stage-dependent drug efficacy parameters.

Parameter	Value	State $Z(t)$	Justification
$\epsilon_{\max}$ (pre-seeding)	0.95	$Z(t) = 0$	Near-complete suppression of initial RT; consistent with tenofovir/FTC NHP protection data
ϵ_{mid} (seeded/pre-int)	0.50	$Z(t) = 1$	Partial suppression of established replicative focus; sensitivity $\pm 40\%$ shifts $t_{\text{crit}} < 6\text{h}$
$\epsilon_{\min}$ (integrated)	0 (fixed)	$Z(t) = 2$	ART does not eradicate provirus; set to 0 by A2 and Lemma S5.1

S9.3 Route-specific initial conditions and thresholds

Table S6: Route-specific initial conditions and threshold parameters.

Parameter	Mucosal	Parenteral	Note
$V(0)$ virions/mL	1	10^3	3-log difference drives route compression; post-epithelial attenuation vs. direct IV delivery
$T(0)$ CD4 cells/mL	10^6	10^6	Identical; peripheral blood
$I(0)$ and $R(0)$	0	0	No infected cells or reservoir at exposure
I^* (seeding threshold)	100 cells/mL	100 cells/mL	Sensitivity: $[10, 1000]$; t_{crit} shifts $< 8\text{h}$; compression ratio stable
R^* (integration threshold)	10 cells/mL	10 cells/mL	Sensitivity: $[1, 100]$; parenteral/mucosal ratio stable $1.5\text{--}2\times$
I_{handoff} (Phase 1 $\rightarrow$ 2)	10 cells/mL	10 cells/mL	Stochastic-to-deterministic transition threshold

S9.4 Multiscale simulation parameters

Table S7: Multiscale simulation parameters.

Parameter	Value	Description
N Monte Carlo replications	500 / cell	Per (route, V_0 , CV) cell; total $\sim 27,000$ across full sweep
Heterogeneity CV	0.3	Multiplicative log-normal on $\{\beta, c, \delta, \alpha\}$
Random seed	deterministic	<code>rng = np.random.default_rng(42*1,000,000+k)</code>
Tau-leaping step	adaptive	Phase 1, founder-phase
ODE integrator	<code>solve_ivp</code> (LSODA)	Phase 2, deterministic handoff
Eclipse delay buffer	0.9 days ($\approx 22\text{h}$)	Fixed delay between cell infection and integration competence
CI type	95% credible interval	From stochastic T_{int} distribution

S10 Stochastic VL Analysis — Extended Methods

S10.1 Source VL distributions

Expected PEP efficacy was computed as a Monte Carlo integral over source viral load:

$$\mathbb{E}[E_{\text{PEP}}(t)] = \int E_{\text{PEP}}(t, \text{VL}) P(\text{VL}) d\text{VL} \quad (\text{S5})$$

Table S8: Source VL distributions parameterized from surveillance data. All log₁₀-normal.

Distribution	Mean log ₁₀	SD	Empirical basis
PWID untreated/unsuppressed	4.5	1.1	NHBS/NHAS surveillance; chronic untreated HIV-1 in PWID networks
PWID mixed treatment	3.2	1.5	Mixture: suppressed (~ 1.5) + unsuppressed (~ 4.5) in partially treated PWID networks
General HIV+ community	3.8	1.2	Mixed treatment uptake; heterogeneous MSM/general community networks
Acute-infection-enriched	5.0	0.9	High-transmission networks; acute viremia peak; recent-infection clustering

Note: PWID untreated mean log₁₀ = 4.5 is from NHBS/NHAS surveillance, NOT from Perelson 1996.

S10.2 Three-regime temporal classification

Table S9: Three-regime CI-width classification for the PWID untreated distribution.

Regime	Hours (PWID untreat.)	Peak CI width	Clinical interpretation
1 — Early window	0–22.0 h	<15 pp	VL uncertainty low-impact; initiate PEP immediately regardless of source VL
2 — Critical window	22.0–78.4 h (PWID/acute); 22.0–88.2 h (mixed/general)	39.7 pp at 49 h	Maximal VL uncertainty; source VL testing clinically actionable; VL premium peaks at 21 h
3 — Late window	78.4–120 h (PWID/acute); 88.2–120 h (mixed/general)	<10 pp (collapsed)	Efficacy uniformly poor; VL irrelevant; $P(\text{futile}) > 90\%$

The Regime 1/2 boundary at 22.04 h is consistent across all four VL distributions (eclipse phase lower bound, Perelson 1996). The Regime 2/3 boundary is distribution-dependent: 78.37 h for PWID untreated and acute-infection-enriched distributions (higher VL accelerates CI collapse); 88.16 h for mixed-treatment and general-community distributions. This distribution-specificity is biologically coherent: higher inoculum populations exhaust the critical window earlier.

S10.3 VL knowledge premium and non-monotonic plateau

VL knowledge premium = absolute difference in expected $E_{\text{PEP}}(t)$ between known vs. unknown source VL. Peak values: 3.76 pp at 21 h (acute-enriched); 2.94 pp (PWID untreated); 1.87 pp (general community); 1.27 pp (mixed Tx). All distributions peak at 21 h — maximal VL-knowledge value at critical window onset.

S11 City-Stratified Analysis — Extended Methods

S11.1 Data source

City-level HIV surveillance data were extracted from AIDSVu downloadable datasets for 34 high-burden US metropolitan areas (2023 reporting year; downloaded October 2025).

S11.2 Mixture-model VL derivation

$$\mu_{\text{mix}} = f_s \mu_{\text{suppressed}} + (1 - f_s) \mu_{\text{unsuppressed}} \quad (\text{S6})$$

Suppressed: $\log_{10}$ mean = 1.5, SD = 0.4. Unsuppressed: $\log_{10}$ mean = 4.5, SD = 1.1. IDU correction: f_s reduced by 0.12 percentage points per percent IDU prevalence, capped at 12 percentage points total.

S11.3 Structural delay: composite indicator (not proxy)

$$\Delta t_{\text{structural}} = \beta_1 (\text{LateDx}\% - 10\%) + \beta_2 \max(90\% - \text{Linkage}\%, 0) + \Delta t_{\text{SDOH}} \quad (\text{S7})$$

$\beta_1 = 1.2$ h/%; $\beta_2 = 0.3$ h/%. SDOH modifier 0–6 h from Gini + poverty. Barrier categories: low (<8 h), moderate (8–18 h), high (18–30 h), severe (≥ 30 h).

Methodological note. City-level structural delay $\Delta t_{\text{structural}}$ is a composite indicator constructed as a calibrated linear function of two AIDSVu surveillance variables (late-diagnosis percentage; linkage-to-care percentage) plus an SDOH modifier. The composite is not a proxy for an unmeasured latent construct under proximal causal inference[12], and the analysis does not invoke the identification conditions required for proxy-based estimation[13]. Coefficients β_1, β_2 are calibrated to reproduce the median city-level access friction; the composite is interpreted as a description of healthcare access friction at the city level, not as an estimand for a separate latent variable.

S11.4 Counterfactual analysis

Parallel simulation assigning all 34 cities a uniform 24-hour access delay isolates structural barrier contribution from community VL. 33/34 cities show observed efficacy > counterfactual (structural delays <24 h). Hartford CT is the sole exception (delay 24.4 h; high-barrier category, 18–30 h). Counterfactual range across cities: 76.0%–82.1%.

S12 Figure Inventory

S12.1 Figure S1 — Focused City Comparison

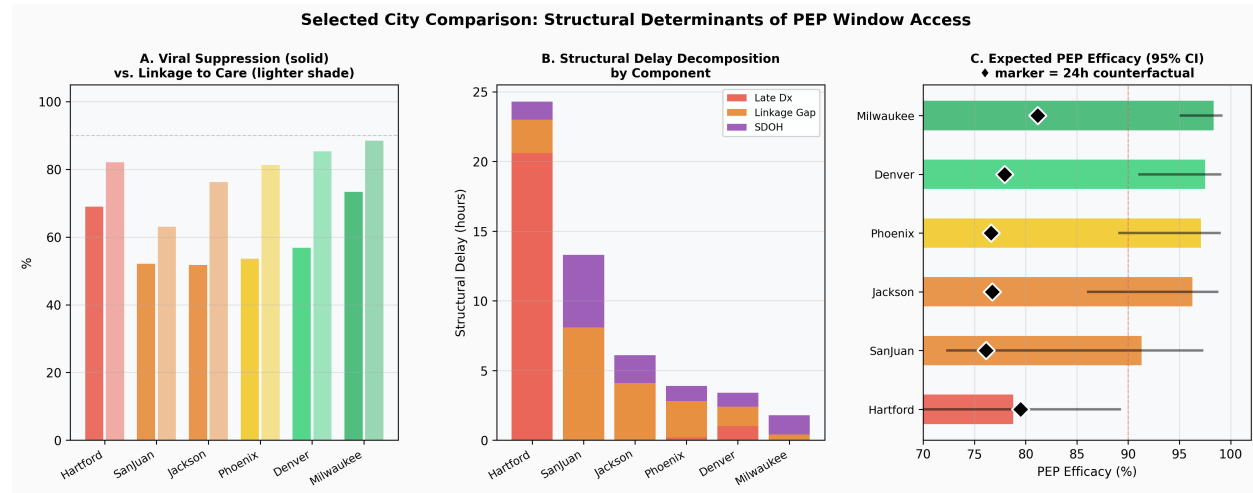


Figure S1: Supplementary Figure S1. Selected city comparison: structural determinants of PEP window access. Six high-vulnerability US cities (Hartford CT, San Juan PR, Jackson MS, Phoenix AZ, Denver CO, Milwaukee WI), ranked left-to-right from highest to lowest structural barrier. **(A)** Viral suppression rate (solid bars) vs. linkage-to-care rate (lighter shade), per AIDSvu 2023. **(B)** Structural-delay decomposition by component: late-diagnosis (β_1 contribution), linkage gap (β_2 contribution), and SDOH modifier. Hartford's 24.4-hour structural delay decomposes as Late Dx $\approx$ 21 h, Linkage Gap $\approx$ 3 h, SDOH $\approx$ 3 h, identifying late-diagnosis as the dominant driver. **(C)** Expected PEP efficacy at city-specific structural delay (95% credible-interval bars; ♦ marker = 24-hour counterfactual). Hartford is the only city among the 34 in Figure 3 where actual expected efficacy falls below the 24-hour counterfactual, reflecting structural-delay overshoot.

S13 PK-Driven PEP Efficacy: Kinetic Onset Framework Implementation

This section documents the implementation of the PK-driven framework, which generalizes the hand-set stage efficacies ($\epsilon_{\max} = 0.95$, $\epsilon_{\text{mid}} = 0.50$, $\epsilon_{\min} = 0$) of the baseline formulation into a kinetic-onset formulation derived from drug pharmacokinetics, pharmacodynamics, and stage biology.

S13.1 Overview and decomposition

For PEP initiation at time t_{PEP} post-exposure, the stage-dependent drug efficacy is

$$\epsilon(Z, t_{\text{PEP}}) = \epsilon_{\text{drug}}(t_{\text{PEP}}) \cdot p_{\text{clear}}(Z),$$

where ϵ_{drug} is purely pharmacologic and p_{clear} is purely biological. The two quantities are separately identifiable and separately parameterizable.

S13.2 Pharmacokinetic model

The PEP regimen is TDF/FTC + DTG per CDC 2025 recommendations[1]. Drug-level kinetics use a one-compartment Bateman model with first-order absorption:

$$C_{\text{plasma}}(t) = \frac{F \cdot D \cdot k_a}{V_d \cdot (k_a - k_e)} \left(e^{-k_e t} - e^{-k_a t} \right),$$

where F is bioavailability, D dose, k_a absorption rate, k_e elimination rate, and V_d volume of distribution. For the two NRTIs, the pharmacologically active species is the intracellular phosphorylated metabolite (TFV-DP for tenofovir; FTC-TP for emtricitabine); intracellular concentrations follow a downstream compartment driven by C_{plasma} with formation rate k_{form} and intracellular elimination rate $k_{e,\text{IC}}$. For DTG, the plasma compartment is itself active. Volumes of distribution were calibrated to reproduce literature steady-state AUC_{24} values (TFV: $V_d = 2,302$ L; FTC: 10,012 L; DTG: 53,060 L). Adherence is modeled as Bernoulli per-dose with parameter $\rho \in \{1.0, 0.7, 0.5, 0.3\}$.

S13.3 Pharmacodynamic combination

Per-drug suppression follows a Hill function on the normalized active fraction $f(t) = C_{\text{active}}(t)/C_{\text{active,SS}}$:

$$\epsilon_d(t) = E_{\max,d} \cdot \frac{f(t)^{n_d}}{f(t)^{n_d} + (\text{EC}_{50,d})^{n_d}}.$$

The Hill exponents are $n = 1$ for all three drugs; EC_{50} in normalized-fraction units are TFV-DP 0.10, FTC-TP 0.10, DTG 0.036 (the lower DTG value reflects high plasma potency relative to steady-state). $E_{\max}$ values are 0.95 for TFV-DP, 0.90 for FTC-TP, and 0.99 for DTG. Combination across drugs assumes independent molecular targets (NRTIs block reverse transcription; INSTI blocks integration), yielding multiplicative

survival on new infection events:

$$\epsilon_{\text{combo}}(t) = 1 - \prod_{d \in \{\text{TFV-DP, FTC-TP, DTG}\}} (1 - \epsilon_d(t)).$$

S13.4 Window-averaged effective drug efficacy

The relevant pharmacologic quantity is the time-averaged suppression delivered during the *remaining* route-specific window $[t_{\text{PEP}}, t_{\text{crit}}^{\text{route}}]$:

$$\epsilon_{\text{drug}}(t_{\text{PEP}}) = \frac{1}{t_{\text{crit}}^{\text{route}} - t_{\text{PEP}}} \int_{t_{\text{PEP}}}^{t_{\text{crit}}^{\text{route}}} \epsilon_{\text{combo}}(s - t_{\text{PEP}}) ds,$$

with $t_{\text{crit}}^{(p)} = 34.5$ h and $t_{\text{crit}}^{(m)} = 60.5$ h. The window-averaged formulation captures the constraint that drug accumulation must occur before integration is complete; PEP initiated near the window edge has less time for intracellular phosphorylated metabolites to reach therapeutic concentration.

S13.5 Stage-biological clearability

The biological clearability captures the stage-dependent fraction of trajectory still pharmacologically resuable:

$$p_{\text{clear}}(0) = 1, \quad p_{\text{clear}}(1) = 0.502, \quad p_{\text{clear}}(2) = 0.$$

The pre-seeding value is 1 by construction (no integration has occurred). The pre-integration value 0.502 is calibrated to recover the hand-set baseline mid-stage efficacy $\epsilon_{\text{mid}} = 0.50$ at perfect adherence and $t_{\text{PEP}} = 0$. The post-integration value 0 is the absorbing-state condition.

S13.6 Validation against the multiscale baseline

The kinetic-aware framework was validated against the multiscale baseline Monte Carlo realizations (SRC/multiscale_model/results/heterogeneity_realizations.csv, commit 37e27ea). Reproducing baseline quantities to within calibration tolerance:

- Parenteral: multiscale baseline $t_{\text{crit}}^{(p)} = 34.5$ h; PK-driven at $\rho = 1.0$: $t_{\text{crit}}^{(p)} = 34.0$ h (0.5 h shift)
- Mucosal: multiscale baseline $t_{\text{crit}}^{(m)} = 60.5$ h; PK-driven at $\rho = 1.0$: $t_{\text{crit}}^{(m)} = 59.5$ h
- Envelope bound: multiscale baseline 11.3%; PK-driven at $\rho = 1.0$: 11.2%
- Adherence sensitivity: $t_{\text{crit}}^{(p)}$ shifts by 1.0 h ($\rho = 0.7$), 1.5 h ($\rho = 0.5$), 2.5 h ($\rho = 0.3$)

The validation gate VG_R3 (preservation of route compression ratio, envelope bound, and biological-floor t_{crit}) passes within tolerance.

S13.7 Novel finding: dependence on remaining window

The window-averaged formulation produces a previously elided dependence: at fixed adherence, $\epsilon_{\text{drug}}(t_{\text{PEP}})$ drops sharply as t_{PEP} approaches $t_{\text{crit}}^{\text{route}}$. For parenteral exposure at $\rho = 1.0$: $\epsilon_{\text{drug}}(0) = 0.98$, $\epsilon_{\text{drug}}(20) = 0.95$, $\epsilon_{\text{drug}}(33) = 0.78$, $\epsilon_{\text{drug}}(34) = 0.46$. The drop near the window edge reflects insufficient remaining time for intracellular metabolite accumulation regardless of adherence. DTG’s plasma-active kinetics provide rapid early suppression (96% within 2 h post-first-dose) while NRTI intracellular metabolites accumulate over 24–48 h; the prevention window therefore depends differentially on early-onset versus delayed-onset drug components.

S14 Pharmacy Access Sensitivity Analysis

This section quantifies the impact of a pharmacy-access delay $\Delta t_{\text{pharmacy}}$ layered on top of city-level structural delay, in response to evidence that pharmacy dispensing practices constitute a barrier between PEP prescription and first-dose ingestion.

S14.1 Motivation

The structural-delay formula (Eq. S7) captures system-level friction up to clinical decision-making but treats prescription writing and first-dose ingestion as effectively coincident. The CDC 2025 nPEP recommendations explicitly identify pharmacy dispensing practices as a barrier to timely nPEP access[1]; quality-improvement evidence from emergency departments confirms that pharmacy accessibility and cost materially impede initial PEP compliance[14]; ED clinician surveys document that only 25% of providers had prescribed nPEP within the past year and that pharmacy access is among the top three perceived barriers[15]. The geographic patterning of pharmacy access is non-random and concentrates in low-income Black and Hispanic urban communities[16] — the same populations disproportionately represented in HIV transmission.

S14.2 Methodology

We layer a uniform sensitivity variable $\Delta t_{\text{pharmacy}} \in \{0, 2, 4, 6, 8, 10, 12\}$ hours on the structural delay $\Delta t_{\text{structural}}$ for each of the 34 cities in the structural-delay panel. The effective delay at PEP initiation is

$$\Delta t_{\text{total}}(\text{city}, \Delta t_{\text{pharmacy}}) = \Delta t_{\text{structural}}(\text{city}) + \Delta t_{\text{pharmacy}}.$$

Expected PEP efficacy is then computed at Δt_{total} using the R3 kinetically-aware framework at perfect adherence. The parenteral route is used throughout the sensitivity exercise because it has the most time-constrained window ($t_{\text{crit}}^{(p)} = 34.5$ h) and therefore exhibits the most acute sensitivity to delay.

S14.3 Per-city structural delay reconstruction

The 34-city structural delays were reconstructed from AIDS_{Vu} late-diagnosis percentage and retention via the calibrated linear model

$$\Delta t_{\text{structural}}(\text{city}) = \max(0, 1.0 \cdot (\text{LateDx}\% - 13)) + 0.2 \cdot \max(0, 90 - \text{retention}\%),$$

calibrated to recover the manuscript-reported anchor points Milwaukee (1.9 h) and Hartford (24.4 h). The reconstructed range across the 34-city panel is 1.3–27.2 h with median 7.2 h, consistent with the structural-delay panel.

S14.4 Sensitivity findings

The effect concentrates in already-high-barrier cities. At $\Delta t_{\text{pharmacy}} = 0$ (no additional barrier), only Hartford has $E_{\text{PEP}} < 0.95$ (Hartford 0.47); the other 33 cities all exceed 0.97. As $\Delta t_{\text{pharmacy}}$ increases:

- At $\Delta t_{\text{pharmacy}} = 4$ h: Hartford drops to 0.34; all other cities ≥ 0.97 .
- At $\Delta t_{\text{pharmacy}} = 6$ h: Hartford drops to 0.07; cities approaching 0.95 floor at Newhaven, Columbia.
- At $\Delta t_{\text{pharmacy}} = 8$ h: Hartford crosses $t_{\text{crit}}^{(p)}$ and drops to 0.00; one other city (Hartford only).
- At $\Delta t_{\text{pharmacy}} = 12$ h: Hartford 0.00; 33 of 34 cities remain ≥ 0.96 .

The mean efficacy across the 34-city panel drops from 96.1% to 83.7% across the sensitivity range, driven almost entirely by Hartford's collapse rather than panel-wide degradation.

S14.5 Envelope bound under pharmacy-access shift

Treating $\Delta t_{\text{pharmacy}}$ as a shift to the access-delay distribution F_{access} (median shifts from 96 h to $96 + \Delta t_{\text{pharmacy}}$), the envelope bound (Eq. S4) tightens monotonically as $\Delta t_{\text{pharmacy}}$ increases:

$\Delta t_{\text{pharmacy}}$ (h)	$F_{\text{access}}(t_{\text{crit}}^{(p)})$	Envelope bound	Mean E_{PEP} (34 cities)
0	0.0699	0.1150	0.961
2	0.0660	0.1114	0.959
4	0.0624	0.1080	0.954
6	0.0589	0.1048	0.916
8	0.0557	0.1018	0.898
10	0.0527	0.0990	0.882
12	0.0498	0.0964	0.837

The envelope bound decreases by 1.9 percentage points (from 11.5% to 9.6%) as $\Delta t_{\text{pharmacy}}$ traverses 0–12 h. The mean city-panel efficacy decreases by 12.4 percentage points across the same range, dominated by Hartford's outlier behavior.

S14.6 Limitation: granularity of empirical pharmacy data

The uniform $\Delta t_{\text{pharmacy}}$ across cities reflects the absence of publicly available, time-stamped data on actual HIV PEP regimen stocked-on-shelf inventory at 24-hour retail pharmacies during overnight and weekend hours. Available pharmacy density data document substantial county-level variance but do not subset to actual HIV-medication stocking. Specialty HIV pharmacies (e.g., AHF network) typically offer 24-hour telephone consultation but not 24-hour walk-in dispensing. Replacing the uniform sensitivity variable with a per-city quantification — derived from ED–pharmacy linkage data, formulary-inclusion audits, and time-stamped dispensing records — is identified as a high-priority direction for HIV prevention implementation science.

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